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Uncertainty sources in UV-Vis spectrophotometric measurement

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Abstract An overview is given of the most important uncertainty sources that affect analytical UV-Vis spectrophotometric measurements. Altogether, eight uncertainty sources are discussed that are expected to have influence in chemical analysis. It is demonstrated that the well-known intrinsic (or “physical”) sources of uncertainty that originate from the instrument itself (repeatability of spectrophotometer reading, spectrophotometer drift, stray light, etc.) often have significantly lower contributions to the combined uncertainty of the result than the “chemical” sources of uncertainty that originate from the object under study (interference from the constituents of the matrix, decomposition of the photometric complex, etc.). Although selectivity

of a photometric procedure is often considered more a validation topic than an uncertainty topic, it is very often important to include it also in the uncertainty budget.

Usually the most difficult part of uncertainty estimation of a chemical measurement result is to evaluate the magnitude of the actual uncertainty components, especially the chemical ones. For most of the uncertainty sources discussed in this paper, approaches for their evaluation are given. A generic uncertainty budget for absorbance is presented.

Keywords UV-Vis spectrophotometry · Measurement uncertainty · Uncertainty sources · Drift · Repeatability · Nonlinearity · Selectivity

Introduction

Uncertainty estimation of chemical analysis results is becoming more and more of a standard requirement. Results without an uncertainty estimate cannot be considered as complete [1, 2].

A common mistake that is done in uncertainty estimation of chemical measurement results via the ISO GUM (The Guide to the Expression of Uncertainty in Measurement) [2] approach is that the well-known “classical” (weighing, volumetric operations, ...) uncertainty sources are well accounted for, while the “chemical” sources of uncertainty, such as decomposition of the analyte, adsorption of the analyte, incomplete selectivity, sample preparation, etc. (see [3] for a good checklist) are often either underestimated or left out of consideration altogether.

Many analytical determinations, including some of the most common (e.g., nitrite [4], phosphorus [5], ammo-

nium [6]) are routinely (in some cases dominantly) carried out spectrophotometrically. The equipment is relatively inexpensive and very rugged. Although atomic spectrometry methods (for elemental analytes) and chromatographic methods (for organic analytes) are competing with spectrophotometry, it is not to be expected that UV-Vis spectrophotometry will cease to be used in chemical analysis.

The physical basis of spectrophotometric measurements is very well understood and the sources of uncertainty originating from the instrument itself (termed as the intrinsic or physical uncertainty sources in this paper) are very well known and have received ample coverage in analytical chemistry literature [7, 8]. There are, however, also numerous sources of uncertainty that originate from the object under study (termed as chemical uncertainty sources in this paper) to spectrophotometric measurements that are more difficult to quantify and have received less attention although they often outweigh the intrinsic ones in their

contribution to the combined uncertainty of the result. What seems to be almost missing from the literature are convenient ways for quantifying such sources of uncertainty.

In this paper, we give an overview of the uncertainty sources – both physical and chemical – encountered in analytical spectrophotometric measurements. In addition to characterizing the uncertainty sources we will in most cases also attempt on the basis of examples to give guidance about how to quantitatively evaluate the respective uncertainty components. Overviews of uncertainty sources have been published for e.g., chromatographic methods [9] but to the best of our knowledge not for UV-Vis spectrophotometry. There have, however, been published quite some uncertainty estimation examples for UV-Vis spectrophotometric measurements, both in the literature [10–12], and on dedicated Web sites [13, 14].

There is no shortage of materials on the propagation of the uncertainties of input quantities of a measurement equation to give the uncertainty of the result (see e.g., [2] and [3]). Therefore, this aspect of uncertainty estimation will not be covered in this paper. Also, the basic concepts and terms of uncertainty calculation (standard uncertainty, coverage factor, etc.) as well as those of spectrophotometry are not defined. Their usage follows the practice set forth in the definitive sources [2, 3] and [8, 15], respectively.

In handling uncertainty estimation clear distinction has to be made between fundamental science measurements and those made at routine analytical laboratories. In a research laboratory it is often possible to devote time to careful study the uncertainty sources and devise ways to correct for those that can be corrected – first of all the systematic effects. This is also the recommended practice in the leading guidance documents [2, 3]. A routine laboratory, however, has usually only very limited resources for such studies and most of the systematic effects have to be included in the uncertainty budget. This paper is meant to be useful from a standpoint of scientific measurements as well as routine analytical determinations.

The discussion, although primarily targeted to the mainstream monochromator spectrophotometers, is sufficiently general, so that it is also applicable to filter photometers and diode array spectrophotometers. Some parts of the discussion are relevant also for the FIA and HPLC instruments but only in very broad terms, because these methods have numerous differences from the conventional UV-Vis spectrophotometry. The conventional absorbance measurements are assumed for quantification. The discussion can also be of use for absorption band area measurements but derivative spectroscopy is excluded.

Uncertainty sources

Below we present an overview of the most important uncertainty sources in analytical spectrophotometric measurements. With the term *analytical spectrophotometric measurements* we refer to measurements of the amount of analyte in an analytical sample using absorption spectrometry in the UV or visible range. The scope of this paper is the

application of spectrophotometry in analytical determinations. Absolute measurements of absorbance or absorptivity values are not considered. The following is noted in compiling the uncertainty sources list:

1. The discussion presented here assumes that the spectrophotometer is in good technical condition and the measurements are performed carefully. Spectrophotometer should be checked regularly. A good checklist is available in [16].
2. We assume that in analytical determinations the calibration graph is constructed using the same spectrometer and the same cells that are used for actual determination.
3. A distinction is made between the intrinsic and chemical uncertainty sources.

These considerations rule out some uncertainty sources, such as the uncertainty of the pathlength of the cells (since absolute absorbance values are not important, the absolute pathlength is also not critical) and imperfect match between the sample and reference cell (is compensated by the calibration graph).

Intrinsic uncertainty sources

Spectrophotometry is in fact a subfield of spectroradiometry. Spectroradiometric measurements – measurements of radiation intensity – are among the least accurate physical measurements [17]. In contrast, analytical spectrophotometric measurements can (in favorable cases) be highly accurate [7]. This is achieved first of all by not measuring absolute radiation intensities but *relative quantities* – transmittance and absorbance. This way many of the uncertainty sources that are important in spectroradiometry will either partially or fully cancel out (drift of the source intensity and detector sensitivity with time, etc.). As a result, combined uncertainties less than 1% are in principle achievable (under favorable conditions) in chemical analysis using spectrophotometric measurements [7].

Absorbance repeatability

According to the VIM (International Vocabulary of Basic and General Terms in Metrology) [18] repeatability is “property of a measuring system to provide closely similar indications for replicated measurements of the same quantity under repeatability conditions.” The most common way of quantitatively expressing repeatability is via the repeatability standard deviation.

Repeatability uncertainty of the spectrophotometer reading in analytical measurements includes the following components: instrumental noise, repeatability of positioning the cell in the cell compartment, possible temperature fluctuations between the solutions and possible dust particles on the cell windows or in the solution [8, 19]. Experimental determination of repeatability uncertainty should be made using repeated measurements involving taking the cell out of the spectrometer and inserting it again

(preferably a spectrum of some other sample should be scanned in between).

Repeatability uncertainty should not be confused with instrumental noise, i.e., the random fluctuations occurring in a signal, which is one of the contributors to the repeatability uncertainty. Uncertainty estimation of an analytical determination involves the overall repeatability of the sample absorbance at a given wavelength, not just the noise.

This is because (1) the analytical results are influenced by the whole repeatability and not only the noise component and (2) modern spectrophotometers employ quite efficient averaging techniques for noise minimization.

In general, the specifications of modern double-beam spectrophotometers allow repeatability in the 0.5 to 1 AU range as “less than 0.000X AU” (See Item 1 of the Electronic Supplementary Material for specific data on some common models) and the repeatability is in broad terms proportional to the absorbance value. The maximum allowed noise (when it is possible to compare) is less by a factor of around two. As a consequence of the Beer’s law the following normally holds in UV-Vis spectrophotometry [7]:

$$\frac{s(C)}{C} = \frac{s(A)}{A} \quad (1)$$

where C is the concentration of the analyte, $s(C)$ is the uncertainty of the analyte concentration, A and $s(A)$ stand for the absorbance of analyte and the uncertainty of analyte absorbance, respectively. This means that the uncertainty contribution to the concentration of the analyte due to the absorbance repeatability according to the instrument specifications is less than $\pm 0.1\%$ (around 0.06% at standard uncertainty level [2]). The most relevant parameters from the Item 1 of the Electronic Supplementary Material are summarized in Table 1.

In Item 2 of the Electronic Supplementary Material there are some datasets of repeatability and drift determination carried out at our laboratory. Expressing and taking repeatability

into account in uncertainty calculations is straightforward – the standard deviation of repeatability is an appropriate quantity to use. The results (after bringing to common way of presenting) were below the values stated in instrument specifications.

The uncertainty due to the intrinsic repeatability of the spectrophotometer has low significance in most chemical analyses or equilibrium constant measurements. In a recent paper [12] from our group comparing photometric procedures based on calibration graph and standard addition it is shown that the joint contribution (uncertainty index) of the repeatabilities of all absorbance measurements (the sample and the calibration graph or standard addition graph points) is only 2.2% and 5.5% of the overall uncertainty in the case of the calibration graph and standard addition method, respectively (see [12] for the calculation of the uncertainty contributions). Similar conclusions can be drawn from the examples of [13, 14].

Spectrophotometer drift

According to the VIM [18], drift is the “change in the indication of a measuring system, generally slow and continuous, related neither to a change in the quantity being measured nor to a change of an influence quantity”. In this paper, we consider only drift of the absorbance scale. Wavelength drift is of minor importance because most measurements are made at the wavelength of the maximum and the maxima of UV-Vis spectra are broad.

In chemical analysis, absorbance drift can be of two types:

1. The intrinsic or “physical” drift, caused by changes of the spectrophotometer with time (e.g., due to changes in ambient temperature during the day), and
2. The “chemical” drift, i.e., drift due to changes occurring with time in the solution under study.

In this section, we discuss only the intrinsic drift. The intrinsic drift can also be divided into two different effects:

Table 1 Parameters from the specifications of some common UV-Vis spectrophotometer models together with data determined experimentally

Parameter	Range	Comments
Baseline stability	± 0.0005 to ± 0.001 AU/h	This parameter is seldom specified
Photometric stability	<0.0003 AU/h	The same value was quoted for all the models that were under examination
Baseline Drift, experimental ^a	≤ 0.0001 to ≤ 0.0005 AU/h	at 600 nm, data for only two models were available
Baseline flatness	± 0.00025 to ± 0.001 AU	
Baseline noise at 1 AU	0.000030 to <0.0002 AU	Usual parameters: 500 nm, 1.5 to 2 nm slit
Noise at 1 AU	<0.0002 to <0.0004 AU	
at 2 AU	<0.0003 AU	This parameter is seldom specified
at 3 AU	0.00014 to <0.0004 AU	This parameter is seldom specified
Photometric at 1 AU	<0.0008 to ± 0.002 AU	Widely differing experimental conditions
repeatability at 2 AU	± 0.004 AU	Widely differing experimental conditions
at 3 AU	± 0.008 AU	Widely differing experimental conditions

^aDetermined experimentally at the laboratory of the authors. Data of only the datasets included in the Electronic Supplementary Material have been used, so the estimates cannot be considered definitive

1. Baseline drift. This effect leads to changes of the intercept of the calibration graph not affecting the spectrometer response (a translational effect according to [20]).
2. Drift of the spectrometer response. This effect leads to change in the spectrometer response not affecting the baseline (a rotational effect according to [20]).

The intrinsic drift of modern spectrophotometers is low and is almost entirely composed of the baseline drift. This is evident from the instrument specifications, even though different manufacturers present drift/stability data differently. The drift is generally stated to be below 0.0003–0.0005 AU/h after warm-up. If the spectrophotometer performs according to specifications, and the baseline is corrected several times during the day and the absorbance values are not extremely low, then the effect of intrinsic drift on analysis result can be made very small.

Drift is a systematic effect, and hence it should be in principle corrected rather than taken into uncertainty budget [2]. Drift can be modeled using different functions, including polynomials, exponential functions, etc. It is, however, impractical in routine analysis to correct for drift because it is usually small and correcting can be work-intensive. Therefore it is usually included in the uncertainty budget. Practical handling of drift as an uncertainty component has important differences compared to repeatability:

1. Repeatabilities of absorbance values of different solutions (e.g., calibration standards and samples) are statistically independent and do not depend on the sequence of measuring or the time that has passed between the measurement of the standards and the samples. The situation is the opposite in the case of drift: the drift components of uncertainties of the solutions are correlated and dependent on the measuring sequence and time between measurements of standards and samples.
2. Repeatability contribution to uncertainty can be efficiently reduced by repeated measurements. The drift contribution cannot.
3. While repeatability is fairly constant from day to day, the drift can be very different on different days.
4. Differently from repeatability, there is no unequivocal way of expressing the drift uncertainty contribution mathematically.

These features make it complicated to quantify the drift component of uncertainty. At our laboratory we have adopted the following approach for quantifying intrinsic drift [12]. Drift is studied on several days using neutral density glass filters (to avoid any possible chemical effects that can arise from use of solutions, see below) having different absorbance values. The spectra of the filters are scanned in groups of 4–6 spectra at 1 to 3-h time intervals. Between scanning spectra the filter is withdrawn from the cell compartment and repositioned. This way the possible cell positioning uncertainty will also be included in the analysis. In order to separate the drift effect from the repeatability effect, the resulting data are treated with the

ANOVA [21] as described in [22]:

$$u_{\text{rep}} = \sqrt{MS_{\text{within}}} \quad (2)$$

$$u_{\text{drift}} = \sqrt{\frac{MS_{\text{between}} - MS_{\text{within}}}{n}} \quad (3)$$

where u_{rep} is the repeatability standard uncertainty, u_{drift} is the standard uncertainty due to spectrophotometer drift, MS_{within} is the within groups mean squares, MS_{between} is the between groups mean squares and n is the number of spectra in the group. This way, components of uncertainty due to repeatability and drift are obtained from the data at standard deviation level and are directly used in further calculations. In addition to the uncertainty estimates, the numbers of degrees of freedom of these estimates are also found:

$$df_{\text{rep}} = (n - 1) \cdot a \quad (4)$$

$$df_{\text{drift}} = a - 1 \quad (5)$$

where df_{rep} is the number of degrees of freedom of u_{rep} , df_{drift} is the number of degrees of freedom of u_{drift} and a is the number of groups. Baseline drift is a rather constant effect with respect to absorbance, but response drift and repeatability are not. Therefore drift should be determined at different absorbance values. Also, drift can be different at different wavelengths, because of the widely different radiation energy reaching the detector at different wavelengths. This is due to the different radiation intensity of the source, different transmittance of the optical system, and different sensitivity of the detector at different wavelengths. Also, in most contemporary spectrophotometers, different radiation sources are used in different wavelength ranges (deuterium lamp in the UV and halogen lamp in the visible region).

Drift can be significantly different on different days, depending on changes in temperature, voltage, etc. On some days it can be almost absent. Therefore, drift should be determined over several days and it is the safest to take the largest u_{drift} value for the uncertainty estimation. It is important to note that this way of expressing drift is different from the one adopted in instrument specifications. In the specifications, it is usually the maximum permissible change of the baseline absorbance per hour, while using this approach we obtain the standard deviation of the variability of the baseline during the time of the experiment.

If the drift uncertainty contribution is assigned to all absorbance values of the measurement run, calibration standards and samples, then some overestimation of uncertainty will result. The reason is that the drift contributions of calibration graph absorbances and sample absorbances are correlated. Correlation can be taken into account when using the general formula of uncertainty propagation [2],

which has also been implemented in some uncertainty estimation software [23], but obtaining the necessary correlation coefficients can be very difficult. We use the following strategy: we use the maximum (4 h) drift values for all measurements, irrespective of their order (thus somewhat overestimating the uncertainty), we neglect the correlation between the drift components of different measurements (also somewhat overestimating the uncertainty). These overestimations will compensate the possible underestimation of uncertainty caused by differences of drift on different days.

The sample datasets of repeatability and drift determination (Item 2 of the Electronic Supplementary Material) reveal that the intrinsic drift is generally a more important uncertainty source than the intrinsic repeatability. The same conclusion is reached when examining the uncertainty budgets in [12–14]. In [12], for example, the joint contribution of the drift components (only the intrinsic drift was considered) is 18.1% of the combined uncertainty of the result in the case of the calibration graph method. Taking into account that 54.2% of the uncertainty is contributed by sample preparation, the contribution of drift to the uncertainty of the measurement procedure itself is 39.5% making drift the single most important uncertainty contributor.

An efficient way to minimize drift contribution on the analysis result is to measure the absorbance values of the standards and samples alternately in a random order during a short time interval. This is however not always possible in practical work.

Non-linearity of the spectrophotometer's absorbance scale

The instrumental non-linearity of the response has its origin in three effects [7, 8]:

1. The stray light.
2. The non-linear character of the detector element used in the spectrophotometer.
3. The finite bandwidth of the radiation passing through the sample.

The stray light Typical stray light specifications of some instruments are available in Item 1 of the Electronic Supplementary Material. Stray light is very sensitive to the wavelength used, increasing sharply with decreasing of wavelength to 200 nm. Such low wavelengths are seldom of interest in analytical work, however. The most common specification at longer wavelengths is expressed as 0.02%T. This is the transmittance of the test solution that ideally should have zero transmittance. When converted to absorbance scale, the effect is 0.0009, 0.003 and 0.008 AU at absorbance values 1.0, 1.5 and 2.0, respectively. This result supports the general knowledge that for accurate quantitative work absorbance values should not significantly exceed 1.0.

The stray light of an instrument tends to increase with deposition of dust particles on the optics of the instrument. Therefore the instrument should be checked regularly [16].

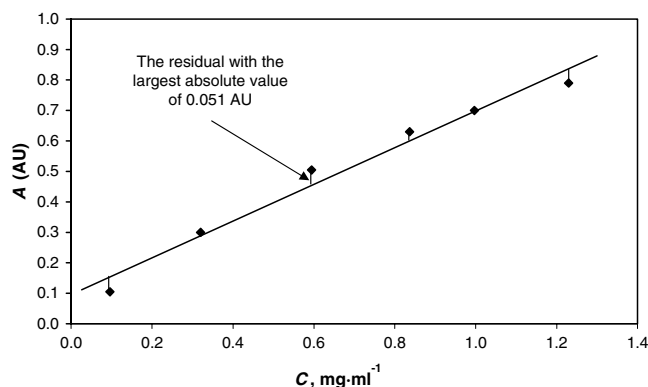


Fig. 1 Handling linearity of calibration graph as an uncertainty source. (a) For the sake of clarity of explanation, the non-linearity of the calibration graph in this figure is greatly exaggerated. In the case of similar highly non-linear response it would certainly be necessary to introduce polynomial or nonlinear calibration

Detector The non-linear character of the used detector element in the spectrophotometer (photomultiplier tube, silicon photodiode, etc.) can also in principle be a source of non-linearity but this effect is very small with modern detectors [17].

The finite bandwidth of radiation. It can be shown [7, 8] that in the case of finite bandwidth non-linearity will be introduced into the calibration graph if the absorbance of the sample differs within the bandwidth. This effect will be minimized if the measurements are carried out at the absorbance maximum.

Non-linearity is another systematic effect that in principle should be corrected rather than taken into account in the uncertainty budget. Again, in cases of small deviations, it is usually impractical at routine laboratories, thus it should be included in the uncertainty budget. We have used the following approach at our laboratory, which is applicable to both intrinsic and chemical non-linearity.

The approach is based on residual analysis. The uncertainty contribution to absorbance due to non-linearity is taken as being “not more than” the absolute value of the largest residual. This value is taken as the estimated uncertainty assuming rectangular distribution. The standard uncertainty contribution is found by dividing by the square root of three [2]. The approach is illustrated in Fig. 1, which presents a case with largely exaggerated non-linearity. A realistic numeric example is given in Item 3 of the Electronic Supplementary Information. The largest residual is 0.00368 AU, leading to a standard uncertainty contribution of 0.0021 AU.

This approach assumes the uncertainty contribution to be constant throughout the calibration curve (thus somewhat overestimating the uncertainty). This is justified because of the properties of the linear regression model [21] – it minimizes the sum of squares of differences of the experimental values from the predicted values – not taking into account in which part of the line the respective data point is located [24]. This constancy has important implications for the uncertainty of the analysis result: it has relatively largest influence in the case of low concentrations. Therefore the

overall calibration line in the case of slight non-linearity is well applicable in the case of intermediate concentrations but not in the case of low concentrations. In the latter case a separate calibration should be carried out including only the points from the beginning of the calibration graph.

Other physical sources of uncertainty

Worth briefly mentioning are the possible partial decomposition of the analyte due to photochemical processes (this can as well be termed as chemical uncertainty source), fluorescence, uncertainty due to rounding of the digital reading and uncertainty due to temperature differences between samples. The first two are of minor significance [7]. The dominating design of UV-Vis spectrophotometers has sample after the monochromator and therefore the radiation energy reaching the sample is low and hardly able to decompose the compounds. Fluorescence radiation is emitted preferably in direction perpendicular to the beam, so that only a very small portion of fluorescent radiation reaches the detector. These days instruments normally have digital readout and are able to give absorbance values with at least 4 digits. This means that rounding of the digital reading has no effect on the uncertainty of the result. The uncertainty due to temperature differences was included in the repeatability component.

Chemical uncertainty sources

Since most analytes cannot be determined by photometric analysis directly – basically because of absence of strong absorption in suitable wavelength range – photometric analysis is very often carried out using a photometric reagent that under certain conditions forms colored complex with the analyte. This is the “weak link” in photometric analysis. There are some well-known criteria of suitability of a particular reaction for use as photometric complex-forming reaction. Ideally the reaction should have the following properties [8]: (a) the reaction between the reagent and analyte should be complete and have exact stoichiometry, which is not altered by other species in the solution; (b) the formed complex should be stable; (c) the photometric reagent itself should have zero absorbance at the analytical wavelength; (d) the reagent should not form complexes with other species present in the solution. In many cases the reagents and reactions are wholly adequate, but very often compromises are necessary.

The chemical uncertainty sources originate from the system under study rather than from the equipment. Systems under study can be very different and with different systems different uncertainty sources dominate. Evaluation of the chemical uncertainty contributions is far more difficult and has received less attention. To the best of our knowledge, there is currently no literature source available that would give an overview about these uncertainty sources.

Some of the sources are not at all limited to spectrophotometric measurements (e.g., instability of the analyte). Nev-

ertheless, we include them here, because their influence on the spectrophotometric measurements is manifested by changes in absorbance values.

In some cases the interfering effects can be minimized by the use of standard addition methods [7, 12], but the most common interference caused by extraneous absorption cannot be overcome this way.

Spectral interferences

This uncertainty source refers to the limited selectivity of UV-Vis spectrophotometric measurement and is one of the most important sources of uncertainty in spectrophotometric measurements. The UV-Vis absorption bands in solution are mostly very broad so that the possibilities of achieving selectivity by wavelength selection are limited. Instead it is usually necessary to separate the compounds of interest from the interfering compounds, mask the interfering compounds and/or shift the absorption maximum of the analyte, usually by complex formation, to a longer wavelength – typically in the visible spectral range (approximately 400–700 nm) – where the other components of the sample do not absorb. Because matrix components can absorb, blank correction is often essential. Worth mentioning are also the various multi-wavelength techniques with multi-linear calibration methods [26] and standard addition methods [12]. Thus there are many possibilities to minimize the interferences – many of them well known for long time [7, 25]. Insufficient attention has been paid to how to estimate the uncertainty contributions from the (residual) interferences [20].

Analysis of real samples generally means analysis of complex mixtures. As in principle every component of the sample affects the result, ideally all these effects should be modeled and taken into account. However, since all the components of the sample are not known to us, this approach can only be used in the case of well-defined objects (e.g., alloys, pharmaceuticals, chemical formulations) for example in industrial process control using multivariate calibration methods [26]. There are usually only one or a few analytes present in the sample to be determined, whereas the other components (the matrix) are of no interest.

Although care is taken while working out an analytical method to ensure that the probable constituents of the matrix do not interfere with determination, there always remains some uncertainty regarding the possible interferents that have not been considered during method development and validation or that have contents exceeding the contents assumed during validation. It is usually desirable to correct for extraneous absorbance, the extent of correction depends on the magnitude of this absorbance and on its effect to the measurement result and accuracy. This can, however involve lengthy studies to find out the causes for the interference.

In many standard methods (e.g., [4, 6]) the interferences have been carefully studied and attached as annexes to the methods. In those annexes it is specified how much will the result of the analysis over or underestimate the analyte

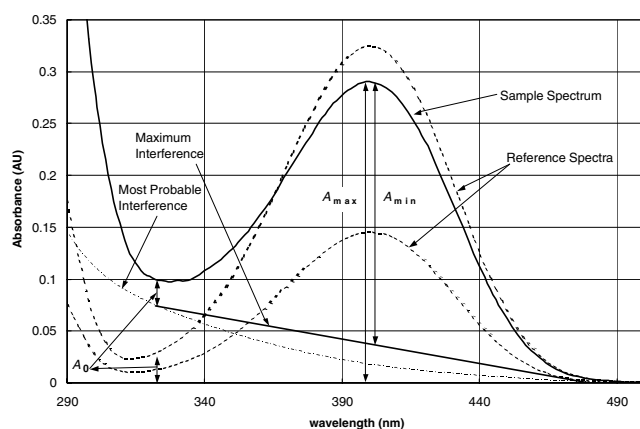


Fig. 2 A possible way of correcting for extraneous absorbance. (a) Solid line sample spectrum; dotted lines standard solution spectra; A_0 is the estimated absorbance “offset” of the sample solution at 322 nm in the case of absence of the interference and it is a necessary parameter for drawing the estimated interference lines. The wavelength is chosen to lie between the wavelengths of the minima of the sample spectrum with interference and those of the undisturbed standard solution spectra. The value of A_0 is estimated from the closest standard solution spectrum. The estimated maximum interference and the estimated most probable interference of the interfering “tail” are denoted by the solid and dash-dot lines, respectively

content if a certain amount of an interferent is present in the sample. Usually the effect is not very large and thus can be introduced into the uncertainty budget as follows. First, the maximum possible content of the interferent in the sample has to be estimated by the analyst as m_{max}^{interf} . Then the estimate of the probable content of the interferent is taken as $m_{probable}^{interf} = m_{max}^{interf}/2$ and its uncertainty is estimated as $\pm m_{max}^{interf}/2$. The effects of these amounts on the analysis result are deduced from the annex. As an example, in determination of nitrite content according to ISO 6777:1984 [4] the 100 μg amount of Fe^{3+} (one of the important interferents) in the sample causes decrease of the result of measuring nitrite nitrogen content from $m_N = 10.0$ to $m_N = 9.49$ μg , i.e., by -0.51 μg [4]. If the maximum possible content of iron in the sample is estimated as, say, $m_{max}^{interf} = 80$ μg , then the $m_{probable}^{interf} = 40$ μg . This probable interference is -0.20 μg and thus the result should be corrected by $+0.20$ μg . The uncertainty of this correction is ± 0.20 μg , which assuming rectangular distribution (the safest) leads to standard uncertainty contribution 0.12 μg . This contribution alone accounts for 1.2% of relative standard uncertainty (if $m_N = 10.0$) which is normally above the combined effect of all the intrinsic uncertainty sources of the spectrophotometer. In addition, there can be several interferents, e.g., in [4] 27 interfering elements or compounds are listed. In more complex samples the combined uncertainty contribution of all the interferents can be substantial.

The most widespread case of such interference is caused by overlapping of the absorption band of the complex with the “tail” of a shorter wavelength absorption band (see the dash-dot line in Fig. 2). A rather useful approach for correcting this interference is presented in Fig. 2.

Approaches like this one are in principle not new [7]. What has not been addressed, however, is how to estimate the uncertainty of such corrections. One way to do this is by the following procedure based on two worst-case scenarios. Since the real absorbance of the interference is unknown, the absorbance at the analytical wavelength is corrected in two ways: underestimating the sample absorbance (A_{min} on Fig. 2), by assuming zero absorbance for the “tail”, and overestimating it (A_{max} on Fig. 2), by assuming that the “tail” is linear (the straight line “Maximum interference”, see the footnote of Fig. 2 for explanations). The best estimate of the sample solution absorbance value $A_{probable}$ that is used for further calculations and its standard uncertainty $u(A_{probable})$ are calculated as follows:

$$A_{probable} = \frac{A_{min} + A_{max}}{2} \quad (6)$$

$$u(A_{probable}) = \frac{A_{max} - A_{min}}{2 \cdot \sqrt{3}} \quad (7)$$

This approach fails if the interferent has absorbance maximum near the wavelength of sample absorbance maximum, but the probability for this is low.

Sometimes it is possible to circumvent the whole problem by correcting the result using the procedural blank. Unfortunately this cannot be taken as a general approach, because very often there is no possibility to obtain the procedural blank that contains the interferents in the same quantities as the sample.

Complexing agent concentration

The amount of the complexing agent added to the solution is an important factor that can affect the measurement [25]. The necessary amount of the complexing agent and the extent by which it can be varied is found during procedure development and/or validation. The amount of the complexing agent is usually well above (very often tens or hundreds of times) the stoichiometric amount. Either too high or too low concentration of the complexing agent can adversely affect the results.

If the concentration of the complexing agent is too low and does not stabilize enough, not all the analyte may be bound into the complex thus leading to lowered absorbance values.

If the amount of the complexing agent is too large then it may introduce spectral interference, as shown in [12]. Normally the reagent has very low (ideally zero) absorbance at the analytical wavelength. At very high complexing agent concentrations other components of the sample can give complex and start interfering with the determination that at lower complexing agent concentrations do not interfere.

The second effect can be minimized if the complexing agent concentration in the calibration solutions and in the sample solution is the same.

It is difficult to give a general advice on quantitatively taking into account this uncertainty component other than estimating its maximum effect during method validation. For example, in a recent study from this laboratory [12] it was found that in determination of iron in aluminum that concentration of the complexing agent (sulfosalicylic acid in this case) in the solution had to be at least ten times higher than the joint concentration of aluminum and iron (on molar scale). This is because the complexing agent at the same time acts as a masking agent for aluminum in this particular determination [25]. Variation from ten to fifteen times did not introduce difference into the results. At lower complexing agent concentrations decreased absorbances were observed.

Composition mismatch between sample solution and calibration solution

This uncertainty source comes into play only in the case of unstable (e.g., Fe^{2+} , Sn^{2+} , etc.) or ill-defined (oil contamination, dietary fiber content, etc.) analytes. The best way to estimate uncertainty in this case is to use the same approach, used above in the case of interference: estimate the maximum possible effect by monitoring the standard solution by establishing the maximum time that the solutions can be kept and evaluating the maximum deviation of the results from those obtained with fresh solutions.

Calibration non-linearity

In addition to the physical uncertainty sources mentioned above, the non-linearity of calibration graph can also originate from chemical reasons (e.g., due to the well-known chromate-dichromate equilibrium [25]). In practical work, there is no need to separate the two – they can be estimated jointly as described above.

Chemical drift

As said above, the chemical drift is manifested by changes of absorbance of the solution because of changes taking place in the solution. The changes can be caused by the following: reactions of matrix compounds between themselves (yielding an absorbing species) or with the complex, reactions of oxygen with the analyte or the complex, simple decomposition of the complex. Unstable complexes are

the reagents, “the color develops during 20 min and the measurement should be carried out during 2 h”.

Experience from our laboratory shows that the color is reasonably stable in simple samples. But even in such samples the decrease of absorbance over 2 h occurs and is in the range of 1% of the absorbance value. In the case of absorbance, of e.g., 0.4, this makes 0.004 AU. This is well above the intrinsic drift of commercial spectrophotometers (see Item 1 of the Electronic Supplementary material).

The situation is still worse in the case of difficult samples like wastewater samples. The decrease of the absorbance during the same time can be around 10–15% of the absorbance value. Of this, 4–6% occurs during the first 20 min. The intrinsic drift of the spectrophotometer (or any other uncertainty source considered above) is nowhere near this. Also, most probably, this effect operates right from the moment of mixing of the reagents and thus the highest observed absorbance value can also be lowered.

Clearly in this case, the most rigorous approach would be to analyze the water thoroughly and try to separate or mask the interfering compounds. This is, however, work-intensive in most cases and often not practical at the routine laboratory level. Therefore, this effect has to be included in the uncertainty budget. Besides the matrix composition, aged or impure reagents can also cause such drift effects. The possible extent of such effects in the matrices within the scope of the intended use should be investigated during method validation.

To estimate the uncertainty contribution due to chemical drift in this particular case of nitrite, we suggest to take the highest absorbance value for the calculation and to estimate its uncertainty as $\pm$ the overall absorbance decrease during 2 h, e.g., $\pm 12\%$. This way, the possible changes that already took place during color formation will also be accounted for. The relative standard uncertainty contribution then will be $\pm 12/\sqrt{3} = 7\%$. This is not an entirely satisfactory approach, but at least the uncertainty estimate of the result will not leave the impression of high precision. The conclusion from the above is that whenever there is a kinetic reaction involved, very large uncertainties (usually far surpassing the other uncertainty components) can be expected.

Discussion

Based on the previous sections, we arrive at the following generic uncertainty budget for the absorbance A of a sample solution:

$$u_c(A) = \sqrt{u_{\text{rep}}^2(A) + u_{\text{drift}}^2(A) + u_{\text{nonlin}}^2(A) + u_{\text{interf}}^2(A) + u_{\text{mismatch}}^2(A) + u_{\text{chem_drift}}^2(A)} \quad (8)$$

not rare in analytical measurements, even in the case of standard methods. Again, a good example is the nitrite determination according to ISO 6777:1984 [4]. This is one of the most common spectrophotometric measurements in routine laboratories. The standard says that after mixing

where the uncertainty components are in the following order: repeatability, intrinsic drift, nonlinearity (both intrinsic and chemical), chemical interferences, sample-calibrant mismatch, and chemical drift. Some of the sources have been omitted because they are not likely to have an

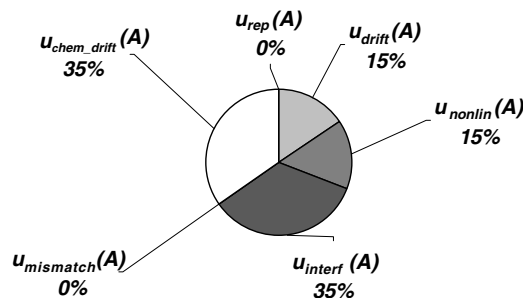


Fig. 3 Uncertainty contributions of different uncertainty components of the absorbance A of a sample solution in the “good” case

influence on analytical spectrophotometric measurement using contemporary equipment, and some are grouped because their contribution is best estimated jointly (see above).

As a very broad generalization for a “good” case under routine laboratory conditions, we can assign the following estimates to these uncertainty components: 0.0001, 0.002, 0.002, 0.003, 0.0, and 0.003 AU (the same order as above, expressed as standard uncertainties, absorbance value $A = 0.40$ AU) (see [12] and the discussion in previous sections). The uncertainty contributions of different uncertainty components for this case are presented in Fig. 3.

These uncertainty components lead to a combined standard uncertainty of 0.005 AU, yielding relative combined standard uncertainty of 1.2% and expanded uncertainty of 2.4% at $k = 2$ level. Thus, even for a well-behaving system, the chemical sources of uncertainty dominate, contributing 69% of the overall uncertainty. The intrinsic sources give the remaining 31% (see [12] for calculation of the percentages).

For comparison, we also give an example of a “complicated case” that is based on the above-discussed nitrite determination. For this case, the estimates of uncertainty components given in Eq. (8) are as follows: 0.0001, 0.002, 0.002, 0.0103, 0.0, and 0.028 AU (the same order as above, expressed as standard uncertainties, absorbance value $A = 0.40$ AU). It can be seen that the numerical values of repeatability, intrinsic drift, non-linearity (both intrinsic and chemical) and sample-calibrant mismatch are the same as those used for the good case. The contribution of chemical interferences was estimated according to the data in Fig. 2 and the chemical drift component was estimated as described earlier. These uncertainty components lead to a combined standard uncertainty of 0.03 AU, yielding relative combined standard uncertainty of 7.5% and expanded uncertainty of 15% at $k = 2$ level. Again, the chemical sources of uncertainty dominate, contributing even 99.1% of the overall uncertainty this case; the intrinsic sources give the remaining 0.9%. The uncertainty contributions of different uncertainty components in this case are shown in Fig. 4.

Skoog et al. [8] in their excellent textbook of instrumental analysis characterize UV-Vis spectrophotometry as a tool for quantitative analysis by saying the following: “typically, relative uncertainties of 1 to 3% are encountered,

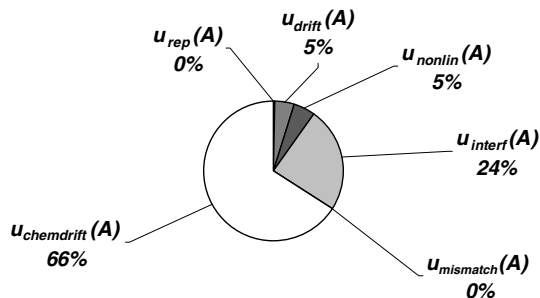


Fig. 4 Uncertainty contributions of different uncertainty components of the absorbance A of a sample solution in a “problematic” case

although with special precautions errors can be reduced to a few tenths of a percent” (p. 342). Based on the generic example above, we can only agree with this statement. If one’s measurements are concerned with objects that do not undergo decomposition, fading of color, extraneous reactions, etc., then the main uncertainty in the measurement arises from the spectrophotometer, and can indeed be few tenths of a percent, if special care is taken. It should be noted, however, that in the case of “normal to difficult” samples (e.g., chemical interference, unstable color, etc.) such low uncertainty is not achievable, whatever the precautions may be. In more common cases, the chemical uncertainty sources dominate, and relative standard uncertainties of the spectrophotometric measurement between 1 and 3% are realistic.

In difficult cases (for example severe chemical drift), however, standard uncertainties in the order of 5–20% are to be expected. For example, in nitrite determination, as discussed above, the lowest achievable combined standard uncertainties could be in the range of ca. 5%, and this would be achievable only for samples with minimum level of interferences.

It is important to keep in mind that all these uncertainty estimates are the uncertainties arising from the spectrophotometric measurement only, and does not include the uncertainty coming from sample preparation (e.g., extraction) or sampling procedures. The latter can (in the case of low analyte contents or difficult (e.g., heterogeneous) samples) contribute more than 90% of the overall uncertainty [27].

Conclusions

Although the physical basis of spectrophotometric measurement is very well developed, estimation of uncertainty of spectrophotometric measurement results is not easy, due to a large number of uncertainty sources, many of which are difficult to quantify. The advances in electronics and materials science has led to spectrophotometers that are highly accurate and the contributions of the intrinsic uncertainty sources to the uncertainty of analytical determinations is usually small. Therefore, dominating importance in uncertainty is held by the chemical uncertainty sources. An overview is given of the uncertainty sources that affect

spectrophotometric analyses. Altogether, eight uncertainty sources are discussed that are expected to have influence in chemical analysis. It is demonstrated that (more or less satisfactorily) most of them can be quantified.

Electronic Supplementary Material

The following material is available as electronic supplementary information:

1. A Table of technical specifications of some common commercial models of UV-Vis spectrophotometers.
2. An MS Excel file containing some sample datasets and their ANOVA analysis of repeatability and drift carried out at our laboratory.
3. An MS Excel file illustrating the handling of nonlinearity of the calibration graph as an uncertainty source.

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